

Demonstration of Proposed Features

Date	1 July 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrate d (Yes / No)	Remarks
1	XGBoost Real-Time Flood Risk Inference	Accepts 10 meteorological indicators (rainfall, humidity, river level, soil moisture, etc.) via a web form, preprocesses them through a fitted StandardScaler, and runs real-time XGBoost classification to return a flood probability percentage and risk status within milliseconds.	Implemented	Yes	Avg inference latency: 78ms. Model accuracy: 97.14%.
2	Multi-Role Authentication & RBAC Guards	Implements a role-based access control system with bcrypt password hashing. Supports four distinct roles — Admin, Meteorologist, Local Authority, and Normal User — each with scoped permissions controlling access to prediction forms, history logs, and model analytics dashboards.	Implemented	Yes	Session-based auth with Flask decorators.
3	Indexed Prediction History & Audit Trail	Logs every prediction transaction (input parameters, risk result, timestamp, user ID) into an indexed SQLite database. Users can retrieve a chronologically sorted, paginated history of all past flood risk evaluations with sub-second query response times.	Implemented	Yes	Database indexes on foreign keys and sort columns.
4	Model Comparison Analytics Dashboard	Admin-only dashboard that visualizes comparative accuracy metrics across Decision Tree	Implemented	Yes	Renders matplotlib-generated

		(94.98%), Random Forest (97.60%), and XGBoost (97.14%) classifiers. Displays ROC curves, feature importance rankings, and model size comparisons to justify the selected classifier.			charts from training artifacts.
5	State-Wise Regional Flood Risk Mapping	Allows users to select from 36 Indian states and union territories for region-specific flood risk evaluation. The prediction context is tagged with geographic metadata, enabling future trend analysis and region-wise historical comparison of flood alerts.	Implemented	Yes	Dropdown with all Indian states in prediction form.
6	Concurrent-Safe WAL Database Engine	Configures SQLite with Write-Ahead Logging (WAL) mode and NORMAL synchronous commits, eliminating file-locking bottlenecks during simultaneous read/write operations. Sustains 73+ requests/second with 0% error rate under 50 concurrent users verified via Locust load testing.	Implemented	Yes	Locust test: 2,027 requests, 0% failures.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%